

# LUMINA PROTOCOL

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## Executive Summary

**Version:** 1.0

**Year:** 2026

**Network:** Base L2

**Website:** <https://www.lumina-org.com>

*For the complete technical whitepaper, visit [lumina-org.com](https://www.lumina-org.com)*

# What is Lumina Protocol?

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Lumina is the first parametric insurance protocol built exclusively for autonomous AI agents in DeFi. Agents purchase coverage via REST API. Claims resolve automatically using Chainlink oracle data. Settlement in USDC. No human intervention required.

## How It Works

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1. Agent owner creates API key + funds agent wallet → 2. Agent calls API to purchase policy → 3. Chainlink monitors trigger conditions → 4. Multisig oracle signs claim → 5. Smart contract pays agent automatically

## Four Products

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| Product           | Covers                  | Premium | Duration    |
|-------------------|-------------------------|---------|-------------|
| Black Swan Shield | Extreme price drops     | 6.5%    | 7-30 days   |
| Depeg Shield      | Stablecoin depegs       | 2.5%    | 14-365 days |
| IL Protection     | Impermanent loss        | 8.5%    | 14-90 days  |
| Exploit Shield    | Smart contract exploits | 4%      | 90-365 days |

## For Liquidity Providers

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Deposit USDC into isolated vaults. Earn yield from agent premiums + Aave V3 lending. Four risk-return profiles (30-365 day cooldowns). Soulbound shares. No deposit/withdrawal fees.

## Revenue Model

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3% protocol fee on every premium. No LP fees. Revenue scales linearly with policy volume.

# Security

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- TimelockController (48h delay) for all admin actions
- Gnosis Safe (2-of-3) multisig governance
- Multisig oracle (2-of-3) — no single point of failure
- 79 automated tests with full coverage
- Separated keys: deployer, oracle, relayer
- Soulbound shares prevent flash loan attacks
- Capital locked at policy purchase — guaranteed solvency

## At a Glance

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| Metric     | Value                       |
|------------|-----------------------------|
| Network    | Base L2 (8453)              |
| Settlement | USDC (real)                 |
| Contracts  | 13 verified on BaseScan     |
| Tests      | 79 passing                  |
| Oracle     | Multisig 2-of-3 + Chainlink |
| Governance | Gnosis Safe + 48h Timelock  |
| Yield      | Premiums + Aave V3          |

## Get Started

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**Website:** [lumina-org.com](https://lumina-org.com)

**GitHub:** [github.com/org-lumina](https://github.com/org-lumina)

**SKILL File:** [lumina-org.com/LUMINA-SKILL.txt](https://lumina-org.com/LUMINA-SKILL.txt)

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